aspect based sentiment analysis with bert indobert

IndoBERT for **Aspect-Based Sentiment Analysis**

Aspect-based sentiment analysis (ABSA) identifies specific target entities and determines their associated sentiment polarities within text (Perwira et al., 2025; Yulianti & Nissa, 2024; Aziz et al., 2024). IndoBERT, a transformer language model pre-trained on large-scale Indonesian text corpora, provides contextual representations that significantly advance Indonesian ABSA tasks beyond traditional machine learning and multilingual baselines (Yulianti & Nissa, 2024; Maretta & Meiriza, 2025; Doewes & Widoyono, 2026).

Does IndoBERT outperform baseline models in Indonesian aspect-based sentiment analysis? N = 14

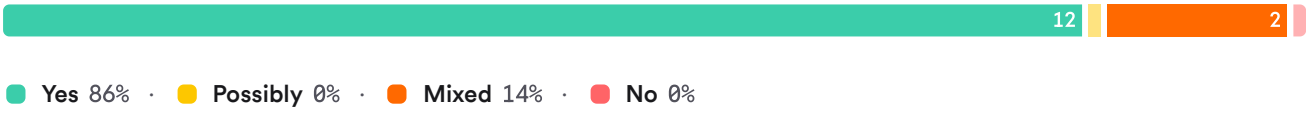


FIGURE 1 Research consensus on IndoBERT performance in Indonesian aspect-based sentiment analysis.

The literature demonstrates strong consensus that fine-tuning IndoBERT consistently outperforms conventional feature-based classifiers, static word embeddings, and multilingual transformer architectures on Indonesian ABSA benchmarks (Yulianti & Nissa, 2024; Aprilah et al., 2025; Doewes & Widoyono, 2026).

IndoBERT Model Architectures and Performance

Researchers implement IndoBERT across diverse architectural paradigms, ranging from sequence tagging and single-sentence classification to sentence-pair modeling using auxiliary sentences (Yulianti & Nissa, 2024; Doewes & Widoyono, 2026). Transforming aspect detection into sentence-pair classification with auxiliary templates yields significant improvements over deep learning baselines by 23.6% and multilingual BERT (mBERT) by 2.2% in F1-score (Yulianti & Nissa, 2024). Similarly, fine-tuning IndoBERT for joint aspect-sentiment extraction and classification matches or surpasses dedicated sequential pipelines (Yulianti & Nissa, 2024; Jazuli et al., 2023; Doewes & Widoyono, 2026).

Application Domain	Dataset Scope	Key Performance Metrics	Benchmark Comparisons
Higher Education Feedback	10,000 university reviews (Jazuli et al., 2024; Jazuli et al., 2023)	Aspect F1: 0.952, Sentiment F1: 0.974 (Jazuli et al., 2024; Jazuli et al., 2023)	Surpasses **prior state-of-the-art** (Jazuli et al., 2024; Jazuli et al., 2023)
Hospitality & Hotels	2,499 luxury hotel reviews (Apriliani et al., 2025)	Accuracy: 95.28%, Macro F1: 82.44% (Apriliani et al., 2025)	Outperforms static **TF-IDF baselines** (Aprilah et al., 2025)
Healthcare Services	2,448 hospital reviews (Maretta & Meiriza, 2025)	Accuracy: 96%, Macro F1: 90% (Maretta & Meiriza, 2025)	Exceeds **SVM and Random Forest**

Application Domain	Dataset Scope	Key Performance Metrics	Benchmark Comparisons
			(Maretta & Meiriza, 2025)
Public Transit & Transport	Access by KAI app reviews (Doewes & Widoyono, 2026)	Aspect Acc: 92.8%, Joint Acc: 96.2% (Doewes & Widoyono, 2026)	Outperforms **mBERT and XLM-R** (Doewes & Widoyono, 2026)
Tourism & Destinations	20,000 Google Reviews (Perwira et al., 2025; Febrianto et al., 2024)	Aspect Precision: 0.85–0.89, Accuracy: 84% (Perwira et al., 2025; Febrianto et al., 2024)	Layer-freezing **optimizes top layers** (Perwira et al., 2025)

FIGURE 2 Performance evaluation across benchmark domains using fine-tuned IndoBERT architectures.

Comparison with General BERT Approaches

Standard BERT architectures rely on bidirectional contextual attention to parse explicit and implicit sentiments across English and non-Indonesian languages (Jazuli et al., 2024; Ahmed et al., 2024; Samia et al., 2022). General BERT frameworks frequently incorporate external syntactic dependency trees, part-of-speech taggers, and knowledge graphs to ground relational aspect-opinion bindings (He et al., 2023; Aziz et al., 2024). For instance, models such as DC-BERT incorporate deep context layers to capture subtle syntactic relations, achieving 92.86% accuracy on standard restaurant datasets (Rajan & Manur, 2024).

In contrast, multilingual variants like mBERT frequently experience domain drift and contextual dilution when applied to low-resource or morphologically rich languages like Indonesian and Hindi (Maretta & Meiriza, 2025; Pathak et al., 2021; Soni, 2025). Monolingual pre-training gives IndoBERT distinct advantages in modeling local slang, regional vocabulary, and colloquial grammar found in user reviews (Perwira et al., 2025; Maretta & Meiriza, 2025; Doewes & Widoyono, 2026). On smartphone review benchmarks, IndoBERT achieved superior accuracies of 96% to 97% across battery, camera, and display aspects, clearly exceeding standard BERT baselines (Setyani et al., 2023).

Methodological Innovations and Limitations

- Multi-label classification with end-to-end IndoBERT captures **inter-aspect relationships** more effectively than traditional single-label pipelines (Mei et al., 2024).
- Integrating topic modeling like **Latent Dirichlet Allocation** with IndoBERT automates unsupervised aspect category discovery before sentiment tagging (Aryanti et al., 2025).
- Severe class imbalance and ambiguous **figurative expressions** consistently degrade performance on neutral and implicit sentiment categories (Perwira et al., 2025; Aprilah et al., 2025; Nur et al., 2026).
- Random oversampling strategies and **weighted loss functions** mitigate sparse representations across minority aspect classes (Aprilah et al., 2025; Maretta & Meiriza, 2025).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Ahmed, S., Samia, M. M., Sayma, M. H., Kabir, M. M., & Mridha, M. F. (2024). tRF-BERT: A transformative approach to aspect-based sentiment analysis in the bengali language. *PLOS ONE, 19*, e0308050 - e0308050. https://doi.org/10.1371/journal.pone.0308050

Aprilah, T., Setiadi, D., & Herowati, W. (2025). Enhancing Aspect-Based Sentiment Analysis via Hugging Face Fine-Tuned IndoBERT. *Journal of Applied Informatics and Computing*. https://doi.org/10.30871/jaic.v9i6.11409

Apriliani, S., Erfina, A., & Warman, C. (2025). Fine-Tuned IndoBERT for Aspect-Based Sentiment Analysis of Indonesian Five-Star Hotel Reviews. *Jurnal Sisfokom (Sistem Informasi dan Komputer)*. https://doi.org/10.32736/sisfokom.v14i4.2491

Aryanti, F., Luthfiarta, A., & Soeroso, D. A. I. (2025). Aspect-Based Sentiment Analysis with LDA and IndoBERT Algorithm on Mental Health App: Riliv. *Journal of Applied Informatics and Computing*. https://doi.org/10.30871/jaic.v9i2.8958

Aziz, K., Ji, D., Chakrabarti, P., Chakrabarti, T., Iqbal, M. S., & Abbasi, R. (2024). Unifying aspect-based sentiment analysis BERT and multi-layered graph convolutional networks for comprehensive sentiment dissection. *Scientific Reports, 14*. https://doi.org/10.1038/s41598-024-61886-7

Doewes, A., & Widoyono, B. (2026). Aspect-Based Sentiment Analysis of Access by KAI Application Reviews Using IndoBERT for Multi-Label Classification Tasks. *Jurnal Teknik Informatika (Jutif)*. https://doi.org/10.52436/1.jutif.2026.7.1.5402

Febrianto, D., Fitriani, M., Afrad, M., & Khadija, M. A. (2024). Aspect Based Sentiment Analysis Menggunakan Indobert Model Terhadap Review Pengunjung Objek Wisata Baturraden. *Melek IT : Information Technology Journal*. https://doi.org/10.30742/melekitjournal.v10i2.358

He, Z., Wang, H., & Zhang, X. (2023). Multi-Task Learning Model Based on BERT and Knowledge Graph for Aspect-Based Sentiment Analysis. *Electronics*. https://doi.org/10.3390/electronics12030737

Jazuli, A., W., & Kusumaningrum, R. (2023). Aspect-based sentiment analysis on student reviews using the Indo-Bert base model. *E3S Web of Conferences*. https://doi.org/10.1051/e3sconf/202344802004

Jazuli, A., W., & Kusumaningrum, R. (2024). Optimizing Aspect-Based Sentiment Analysis Using BERT for Comprehensive Analysis of Indonesian Student Feedback. *Applied Sciences*. https://doi.org/10.3390/app15010172

Maretta, A., & Meiriza, A. (2025). Aspect-Based Sentiment Analysis of Hospital Service Reviews Using Fine-Tuned IndoBERT. *Journal of Applied Informatics and Computing*. https://doi.org/10.30871/jaic.v9i5.10765

Mei, N.-C., Tiun, S., & Sastria, G. (2024). Multi-Label Aspect-Sentiment Classification on Indonesian Cosmetic Product Reviews with IndoBERT Model. *International Journal of Advanced Computer Science and Applications*. https://doi.org/10.14569/ijacsa.2024.0151168

Nur, S., Sahibu, S., & Razak, M. (2026). Aspect-Based Sentiment Analysis of Tourist Attractions in Labuanbajo Using the Transformer Model as a Recommendation for Improving Service Quality. *Journal of Applied Informatics and Computing*. https://doi.org/10.30871/jaic.v10i1.11565

Pathak, A., Kumar, S., Roy, P., & Kim, B.-G. (2021). Aspect-Based Sentiment Analysis in Hindi Language by Ensembling Pre-Trained mBERT Models. *Electronics*. https://doi.org/10.3390/electronics10212641

Perwira, R., Permadi, V. A., Purnamasari, D., & Agusdin, R. P. (2025). Domain-Specific Fine-Tuning of IndoBERT for Aspect-Based Sentiment Analysis in Indonesian Travel User-Generated Content. *Journal of Information Systems Engineering and Business Intelligence*. https://doi.org/10.20473/jisebi.11.1.30-40

Rajan, A., & Manur, M. (2024). Aspect based sentiment analysis using fine-tuned BERT model with deep context features. *IAES International Journal of Artificial Intelligence (IJ-AI)*. https://doi.org/10.11591/ijai.v13.i2.pp1250-1261

Samia, M. M., Rajee, A., Hasan, M. R., Faruq, M., & Paul, P. (2022). Aspect-based Sentiment Analysis for Bengali Text using Bidirectional Encoder Representations from Transformers (BERT). *International Journal of Advanced Computer Science and Applications*. https://doi.org/10.14569/ijacsa.2022.01312112

Setyani, S., Prasetiyowati, S., & Sibaroni, Y. (2023). Multi Aspect Sentiment Analysis of Mutual Funds Investment App Bibit Using BERT Method. *International Journal on Information and Communication Technology (IJoICT)*. https://doi.org/10.21108/ijoict.v9i1.718

Soni, V. K. (2025). Enhancing Aspect Based Sentiment Analysis with a Hybrid Model for Hindi Language. *Journal of Information Systems Engineering and Management*. https://doi.org/10.52783/jisem.v10i20s.3129

Yulianti, E., & Nissa, N. K. (2024). ABSA of Indonesian customer reviews using IndoBERT: single- sentence and sentence-pair classification approaches. *Bulletin of Electrical Engineering and Informatics*. https://doi.org/10.11591/eei.v13i5.8032